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Topic : Linux and shell scripting

Assignment 1: Linux Basic Commands

Section A - Navigation (Practical)

1. Create a directory named linux_practice.
2. Go inside that directory.
3. Create another directory called test_folder.
4. Move back to the parent directory.
5. Display your current working directory.
6. List all files and directories.
7. Find help for the ls command.

```
shinu@Shinu:/mnt/c/Users/shrin/desktop/nucleusteqtraining$ mkdir linux_practice
shinu@Shinu:/mnt/c/Users/shrin/desktop/nucleusteqtraining$ ls
linux_practice
shinu@Shinu:/mnt/c/Users/shrin/desktop/nucleusteqtraining$ cd linux_practice
shinu@Shinu:/mnt/c/Users/shrin/desktop/nucleusteqtraining/linux_practice$ mkdir test_folder
shinu@Shinu:/mnt/c/Users/shrin/desktop/nucleusteqtraining/linux_practice$ ls
test_folder
shinu@Shinu:/mnt/c/Users/shrin/desktop/nucleusteqtraining/linux_practice$ cd ..
shinu@Shinu:/mnt/c/Users/shrin/desktop/nucleusteqtraining$ pwd
/mnt/c/Users/shrin/desktop/nucleusteqtraining
shinu@Shinu:/mnt/c/Users/shrin/desktop/nucleusteqtraining$ ls
linux_practice
shinu@Shinu:/mnt/c/Users/shrin/desktop/nucleusteqtraining$ ls -l
total 0
drwxrwxrwx 1 shinu shinu 4096 Mar 13 14:41 linux_practice
```

```
shinu@Shinu:/mnt/c/Users/shrin/desktop/nucleusteqtraining$ ls --help
Usage: ls [OPTION]... [FILE]...
List information about the FILEs (the current directory by default).
Sort entries alphabetically if none of -cftuvSUX nor --sort is specified.

Mandatory arguments to long options are mandatory for short options too.
  -a, --all                  do not ignore entries starting with .
  -A, --almost-all           do not list implied . and ..
      --author               with -l, print the author of each file
  -b, --escape               print C-style escapes for nongraphic characters
      --block-size=SIZE      with -l, scale sizes by SIZE when printing them;
                               e.g., '--block-size=M'; see SIZE format below
  -B, --ignore-backups       do not list implied entries ending with ~
  -c                         with -lt: sort by, and show, ctime (time of last
                               modification of file status information);
                               with -l: show ctime and sort by name;
                               otherwise: sort by ctime, newest first
  -C                         list entries by columns
      --color[=WHEN]         colorize the output; WHEN can be 'always' (default
                               if omitted), 'auto', or 'never'; more info below
  -d, --directory            list directories themselves, not their contents
  -D, --dired                generate output designed for Emacs' dired mode
  -f                         do not sort, enable -aU, disable -ls --color
  -F, --classify             append indicator (one of */=>@|) to entries
      --file-type            likewise, except do not append '*'
      --format=WORD          across -x, commas -m, horizontal -x, long -l,
                               single-column -1, verbose -l, vertical -C
      --full-time            like -l --time-style=full-iso
  -g                         like -l, but do not list owner
      --group-directories-first
                             group directories before files;
                               can be augmented with a --sort option, but any
                               use of --sort=none (-U) disables grouping
  -G, --no-group             in a long listing, don't print group names
  -h, --human-readable       with -l and -s, print sizes like 1K 234M 2G etc.
      --si                   likewise, but use powers of 1000 not 1024
  -H, --dereference-command-line
                             follow symbolic links listed on the command line
```

```
shinu@Shinu:/mnt/c/Users/shrin/desktop/nucleusteqtraining$ tree
.
└── linux_practice
    └── test_folder

2 directories, 0 files
shinu@Shinu:/mnt/c/Users/shrin/desktop/nucleusteqtraining$
```

Section B - File & Directory Operations

1. Create a file named student.txt.
2. Create a directory called data.
3. Copy student.txt inside data.
4. Rename student.txt to info.txt.
5. Delete the file info.txt.
6. Remove the data directory using:
    - rmdir
    - rm -rf

```
shinu@Shinu:/mnt/c/Users/shrin/desktop/nucleusteqtraining$ touch student.txt
shinu@Shinu:/mnt/c/Users/shrin/desktop/nucleusteqtraining$ ls
student.txt
shinu@Shinu:/mnt/c/Users/shrin/desktop/nucleusteqtraining$ echo "Name : Shrinivas" >> student.txt
shinu@Shinu:/mnt/c/Users/shrin/desktop/nucleusteqtraining$ echo "College : NIT AP" >> student.txt
shinu@Shinu:/mnt/c/Users/shrin/desktop/nucleusteqtraining$ echo "Batch : 2026" >> student.txt
shinu@Shinu:/mnt/c/Users/shrin/desktop/nucleusteqtraining$ cat student.txt
Name : Shrinivas
College : NIT AP
Batch : 2026
shinu@Shinu:/mnt/c/Users/shrin/desktop/nucleusteqtraining$ mkdir data
shinu@Shinu:/mnt/c/Users/shrin/desktop/nucleusteqtraining$ ls
data  student.txt
shinu@Shinu:/mnt/c/Users/shrin/desktop/nucleusteqtraining$ cp student.txt data
shinu@Shinu:/mnt/c/Users/shrin/desktop/nucleusteqtraining$ tree
.
├── data
│   └── student.txt
└── student.txt

1 directory, 2 files
shinu@Shinu:/mnt/c/Users/shrin/desktop/nucleusteqtraining$ mv student.txt info.txt
shinu@Shinu:/mnt/c/Users/shrin/desktop/nucleusteqtraining$ ls
data  info.txt
shinu@Shinu:/mnt/c/Users/shrin/desktop/nucleusteqtraining$ cat info.txt
Name : Shrinivas
College : NIT AP
Batch : 2026
shinu@Shinu:/mnt/c/Users/shrin/desktop/nucleusteqtraining$ rm info.txt
shinu@Shinu:/mnt/c/Users/shrin/desktop/nucleusteqtraining$ ls
data
```

rmdir removes only empty directories

Since data contains files, it cannot remove it

```
shinu@Shinu:/mnt/c/Users/shrin/desktop/nucleusteqtraining$ rmdir data
rmdir: failed to remove 'data': Directory not empty
shinu@Shinu:/mnt/c/Users/shrin/desktop/nucleusteqtraining$ tree
.
└── data
    └── student.txt

1 directory, 1 file
shinu@Shinu:/mnt/c/Users/shrin/desktop/nucleusteqtraining$ rm -rf data
shinu@Shinu:/mnt/c/Users/shrin/desktop/nucleusteqtraining$ ls
```

Assignment 2: File Editing & TextProcessing

1. Create a file called bio.txt.

2. Add the following details using vi:

- Your Name
- Course Name
- City

3. Save the file properly.

```
shinu@Shinu:/mnt/c/Users/shrin/desktop/nucleusteqtraining$ touch bio.txt
shinu@Shinu:/mnt/c/Users/shrin/desktop/nucleusteqtraining$ vi bio.txt
shinu@Shinu:/mnt/c/Users/shrin/desktop/nucleusteqtraining$ cat bio.txt
Shrinivas
Linux Training
Nanded Maharashtra
```

4. Search for your name inside the file.

```
Shrinivas
Linux Training
Nanded Maharashtra
~
~
~
~
~
~
~
~
~
~
~
~
~
~
~
~
~
~
~
~
~
~
~
~
~
~
~
~
~
~
~
~
~
~
~
~
~
/Shrinivas
```

5. Delete all content from the file using shortcut commands.

Esc → Switch to command mode in Vim.

gg → Move the cursor to the top of the file.

VG → Select the entire file from top to bottom.

d → Delete everything that is selected.So when used together:

Esc → gg → VG → d

It goes to the top of the file and removes all the text.

```
shinu@Shinu:/mnt/c/Users/shrin/desktop/nucleusteqtraining$ vi bio.txt
shinu@Shinu:/mnt/c/Users/shrin/desktop/nucleusteqtraining$ cat bio.txt
shinu@Shinu:/mnt/c/Users/shrin/desktop/nucleusteqtraining$
```

6. Use grep to:

1) Search your name in bio.txt
2) Show lines that DO NOT contain your name.

```
shinu@Shinu:/mnt/c/Users/shrin/desktop/nucleusteqtraining$ cat bio.txt | grep "Shrinivas"
Shrinivas
shinu@Shinu:/mnt/c/Users/shrin/desktop/nucleusteqtraining$ cat bio.txt | grep -v "Shrinivas"
Linux Training
Nanded Maharashtra
shinu@Shinu:/mnt/c/Users/shrin/desktop/nucleusteqtraining$
```

Assignment 3: Linux Permissions

1. Create a file script.sh.

2. Check its permission using ls -l.

```
shinu@Shinu:~$ touch script.sh
shinu@Shinu:~$ ls -l
total 713020
drwxr-xr-x 11 shinu shinu      4096 Aug 15  2025 hadoop
-rw-r--r--  1 shinu shinu 730107476 Jun 26  2023 hadoop-3.3.6.tar.gz
drwxrwxr-x  2 shinu shinu      4096 Aug 15  2025 hadoop_input
drwxr-xr-x  3 shinu shinu      4096 Aug 16  2025 hadoop_project
drwxrwxr-x  4 shinu shinu      4096 Aug 15  2025 hadoopdata
-rw-r--r--  1 shinu shinu         0 Mar 13 15:51 script.sh
drwxr-xr-x  3 shinu shinu      4096 Aug 16  2025 sgpa
```

3. Change permission to:

i) 755

```
shinu@Shinu:~$ chmod 755 script.sh
shinu@Shinu:~$ ls -l
total 713020
drwxr-xr-x 11 shinu shinu      4096 Aug 15  2025 hadoop
-rw-r--r--  1 shinu shinu 730107476 Jun 26  2023 hadoop-3.3.6.tar.gz
drwxrwxr-x  2 shinu shinu      4096 Aug 15  2025 hadoop_input
drwxr-xr-x  3 shinu shinu      4096 Aug 16  2025 hadoop_project
drwxrwxr-x  4 shinu shinu      4096 Aug 15  2025 hadoopdata
-rwxr-xr-x  1 shinu shinu         0 Mar 13 15:51 script.sh
drwxr-xr-x  3 shinu shinu      4096 Aug 16  2025 sgpa
```

ii) 764

```
shinu@Shinu:~$ chmod 764 script.sh
shinu@Shinu:~$ ls -l
total 713020
drwxr-xr-x 11 shinu shinu      4096 Aug 15  2025 hadoop
-rw-r--r--  1 shinu shinu 730107476 Jun 26  2023 hadoop-3.3.6.tar.gz
drwxrwxr-x  2 shinu shinu      4096 Aug 15  2025 hadoop_input
drwxr-xr-x  3 shinu shinu      4096 Aug 16  2025 hadoop_project
drwxrwxr-x  4 shinu shinu      4096 Aug 15  2025 hadoopdata
-rwxrw-r--  1 shinu shinu         0 Mar 13 15:51 script.sh
drwxr-xr-x  3 shinu shinu      4096 Aug 16  2025 sgpa
```

4. Explain what 764 means.

First number shows permission for owner

Second number shows permission for group

Third number shows permission for others

Owner = 7 = 4 (read) + 2 (write) + 1 (execute) i.e. Owner can read , write and execute

Group = 6 = 4 (read) + 2 (write) i.e. Group can read and write

Others = 4 = 4 (read) i.e. Others can only read file

5. Give execute permission using symbolic method (chmod +x).

```
shinu@Shinu:~$ chmod +x script.sh
shinu@Shinu:~$ ls -l
total 713020
drwxr-xr-x 11 shinu shinu      4096 Aug 15  2025 hadoop
-rw-r--r--  1 shinu shinu 730107476 Jun 26  2023 hadoop-3.3.6.tar.gz
drwxrwxr-x  2 shinu shinu      4096 Aug 15  2025 hadoop_input
drwxr-xr-x  3 shinu shinu      4096 Aug 16  2025 hadoop_project
drwxrwxr-x  4 shinu shinu      4096 Aug 15  2025 hadoopdata
-rwxrwxr-x  1 shinu shinu         0 Mar 13 15:51 script.sh
drwxr-xr-x  3 shinu shinu      4096 Aug 16  2025 sgpa
```

6. What is the meaning of r=4, w=2, x=1?

r = 4 - read permission

w = 2 - write permission

x = 1 - execute permission

7. What does permission 777 mean? Is it safe? Why?

Permission 777 means full access to everyone.

Owner = 7 : read, write, execute

Group = 7 : read, write, execute

Others = 7 : read, write, execute

So anyone can view, change, and run the file.

No, it is usually not safe.

Because every user on the system can modify or delete the file. This can cause security problems.

Assignment 4 Linux Networking

Task : Check Network Configuration

1. Check your system IP address.

```
shinu@Shinu:/mnt/c/Users/shrin$ ip addr
1: lo: <LOOPBACK,UP,LOWER_UP> mtu 65536 qdisc noqueue state UNKNOWN group default qlen 1000
    link/loopback 00:00:00:00:00:00 brd 00:00:00:00:00:00
    inet 127.0.0.1/8 scope host lo
       valid_lft forever preferred_lft forever
    inet 10.255.255.254/32 brd 10.255.255.254 scope global lo
       valid_lft forever preferred_lft forever
    inet6 ::1/128 scope host
       valid_lft forever preferred_lft forever
2: eth0: <BROADCAST,MULTICAST,UP,LOWER_UP> mtu 1500 qdisc mq state UP group default qlen 1000
    link/ether 00:15:5d:26:ad:f5 brd ff:ff:ff:ff:ff:ff
    inet 172.24.221.125/20 brd 172.24.223.255 scope global eth0
       valid_lft forever preferred_lft forever
    inet6 fe80::215:5dff:fe26:adf5/64 scope link
       valid_lft forever preferred_lft forever
shinu@Shinu:/mnt/c/Users/shrin$ hostname -I
172.24.221.125
```

2. Write down about:

IP Address(Internet Protocol Address)

An IP address is a unique number given to a device when it connects to a network.

It helps devices find and communicate with each other on the internet or within the same network.

Task : Test Network Reachability

3. Ping google.com

```
shinu@Shinu:/mnt/c/Users/shrin$ ping google.com
PING google.com (142.250.206.78) 56(84) bytes of data.
64 bytes from pnmaaa-ay-in-f14.1e100.net (142.250.206.78): icmp_seq=1 ttl=116 time=30.3 ms
64 bytes from pnmaaa-ay-in-f14.1e100.net (142.250.206.78): icmp_seq=2 ttl=116 time=30.6 ms
64 bytes from pnmaaa-ay-in-f14.1e100.net (142.250.206.78): icmp_seq=3 ttl=116 time=30.4 ms
64 bytes from pnmaaa-ay-in-f14.1e100.net (142.250.206.78): icmp_seq=4 ttl=116 time=30.6 ms
64 bytes from pnmaaa-ay-in-f14.1e100.net (142.250.206.78): icmp_seq=5 ttl=116 time=30.1 ms
64 bytes from pnmaaa-ay-in-f14.1e100.net (142.250.206.78): icmp_seq=6 ttl=116 time=30.1 ms
64 bytes from pnmaaa-ay-in-f14.1e100.net (142.250.206.78): icmp_seq=7 ttl=116 time=30.0 ms
64 bytes from pnmaaa-ay-in-f14.1e100.net (142.250.206.78): icmp_seq=8 ttl=116 time=40.8 ms
64 bytes from pnmaaa-ay-in-f14.1e100.net (142.250.206.78): icmp_seq=9 ttl=116 time=33.3 ms
64 bytes from pnmaaa-ay-in-f14.1e100.net (142.250.206.78): icmp_seq=10 ttl=116 time=29.8 ms
64 bytes from pnmaaa-ay-in-f14.1e100.net (142.250.206.78): icmp_seq=11 ttl=116 time=29.4 ms
^C
--- google.com ping statistics ---
11 packets transmitted, 11 received, 0% packet loss, time 9992ms
rtt min/avg/max/mdev = 29.431/31.398/40.833/3.129 ms
```

In below screenshot only 4 packets were sent

```
shinu@Shinu:/mnt/c/Users/shrin$ ping -c 4 google.com
PING google.com (142.250.206.78) 56(84) bytes of data.
64 bytes from pnmaaa-ay-in-f14.1e100.net (142.250.206.78): icmp_seq=1 ttl=116 time=30.0 ms
64 bytes from pnmaaa-ay-in-f14.1e100.net (142.250.206.78): icmp_seq=2 ttl=116 time=30.5 ms
64 bytes from pnmaaa-ay-in-f14.1e100.net (142.250.206.78): icmp_seq=3 ttl=116 time=29.8 ms
64 bytes from pnmaaa-ay-in-f14.1e100.net (142.250.206.78): icmp_seq=4 ttl=116 time=36.0 ms

--- google.com ping statistics ---
4 packets transmitted, 4 received, 0% packet loss, time 3000ms
rtt min/avg/max/mdev = 29.813/31.565/35.975/2.556 ms
```

4. Observe and write:

○ IP address resolved

The system resolved the domain name google.com to the IP address:
142.250.206.78

○ Packet loss %

Packet loss is 0%, which means all the packets sent from the system were successfully received back.

Task : Check Port Connectivity

5. Check if port 80 is open on google.com using telnet:

```
shinu@Shinu:/mnt/c/Users/shrin$ telnet google.com 80
Trying 142.250.206.78...
Connected to google.com.
Escape character is '^]'.
```

6. What does it mean if: Download google.com homepage.

```
shinu@Shinu:/mnt/c/Users/shrin$ telnet google.com 80
Trying 142.251.43.142...
Connected to google.com.
Escape character is '^]'.
GET / HTTP/1.1
Host: google.com

HTTP/1.1 301 Moved Permanently
Location: http://www.google.com/
Content-Type: text/html; charset=UTF-8
Content-Security-Policy-Report-Only: object-src 'none';base-uri 'self';script-src 'nonce-NVssCM09Lum5UQxwlEeQgQ' 'strict
-dynamic' 'report-sample' 'unsafe-eval' 'unsafe-inline' https: http:;report-uri https://csp.withgoogle.com/csp/gws/other
-hp
Reporting-Endpoints: default="//www.google.com/httpservice/retry/jserror?ei=sDW1aYmdJtDA5OUP86D-iAs&cad=crash&error=Page
%20Crash&jsel=1&bver=2397&dpf=QAF50LPxg9N6o6exh4Rn-pBcg-S0V3rznMXmdG0yWjs"
Date: Sat, 14 Mar 2026 10:17:20 GMT
Expires: Mon, 13 Apr 2026 10:17:20 GMT
Cache-Control: public, max-age=2592000
Server: gws
Content-Length: 219
X-XSS-Protection: 0
X-Frame-Options: SAMEORIGIN

<HTML><HEAD><meta http-equiv="content-type" content="text/html;charset=utf-8">
<TITLE>301 Moved</TITLE></HEAD><BODY>
<H1>301 Moved</H1>
The document has moved
<A HREF="http://www.google.com/">here</A>.
</BODY></HTML>
```

7. What does it mean if:

○ Connection successful?

It means the system was able to connect to port 80 on google.com.

This shows that the port is open and the server is accepting connections.

○ Connection refused?

It means the server is reachable but port 80 is closed or not accepting connections.

○ Connection timed out?

It means the system could not get a response from the server.

This usually happens due to network issues.

Task : Download Data Using curl

8. Save it into a file.

```
shinu@Shinu:~$ ls -l
total 713020
drwxr-xr-x 11 shinu shinu      4096 Aug 15  2025 hadoop
-rw-r--r--  1 shinu shinu 730107476 Jun 26  2023 hadoop-3.3.6.tar.gz
drwxrwxr-x  2 shinu shinu      4096 Aug 15  2025 hadoop_input
drwxr-xr-x  3 shinu shinu      4096 Aug 16  2025 hadoop_project
drwxrwxr-x  4 shinu shinu      4096 Aug 15  2025 hadoopdata
-rwxrwxr-x  1 shinu shinu         0 Mar 13 15:51 script.sh
drwxr-xr-x  3 shinu shinu      4096 Aug 16  2025 sgpa
shinu@Shinu:~$ curl google.com >> google.txt
  % Total    % Received % Xferd  Average Speed   Time    Time     Time  Current
                                 Dload  Upload   Total   Spent    Left  Speed
100   219  100   219    0     0    479      0 --:--:-- --:--:-- --:--:--   480
shinu@Shinu:~$ cat google.txt
<HTML><HEAD><meta http-equiv="content-type" content="text/html;charset=utf-8">
<TITLE>301 Moved</TITLE></HEAD><BODY>
<H1>301 Moved</H1>
The document has moved
<A HREF="http://www.google.com/">here</A>.
</BODY></HTML>
```

9. Verify the file is created.

```
shinu@Shinu:~$ ls -l
total 713024
-rw-r--r--  1 shinu shinu       219 Mar 14 16:03 google.txt
drwxr-xr-x 11 shinu shinu      4096 Aug 15  2025 hadoop
-rw-r--r--  1 shinu shinu 730107476 Jun 26  2023 hadoop-3.3.6.tar.gz
drwxrwxr-x  2 shinu shinu      4096 Aug 15  2025 hadoop_input
drwxr-xr-x  3 shinu shinu      4096 Aug 16  2025 hadoop_project
drwxrwxr-x  4 shinu shinu      4096 Aug 15  2025 hadoopdata
-rwxrwxr-x  1 shinu shinu         0 Mar 13 15:51 script.sh
drwxr-xr-x  3 shinu shinu      4096 Aug 16  2025 sgpa
```

Assignment 5: Process & System Monitoring

1. List all running processes.

```
shinu@Shinu:~$ ps aux
USER         PID %CPU %MEM    VSZ   RSS TTY      STAT START   TIME COMMAND
root           1  0.0  0.1 165764 11228 ?        Ss   15:17   0:00 /sbin/init
root           2  0.0  0.0   2476  1432 ?        Sl   15:17   0:00 /init
root           6  0.0  0.0   2492   132 ?        Sl   15:17   0:00 plan9 --control-socket 6 --log-level 4 --server-fd 7 --pipe-fd 9 --log-truncate
root          62  0.0  0.1  64132 14532 ?        S<s  15:17   0:00 /lib/systemd/systemd-journald
root          82  0.0  0.0  21960  5660 ?        Ss   15:17   0:00 /lib/systemd/systemd-udevd
systemd+     131  0.0  0.1  26200 13456 ?        Ss   15:17   0:00 /lib/systemd/systemd-resolved
systemd+     135  0.0  0.0  89364  6684 ?        Ssl  15:17   0:00 /lib/systemd/systemd-timesyncd
root         178  0.0  0.0   4308  2744 ?        Ss   15:17   0:00 /usr/sbin/cron -f -P
message+     180  0.0  0.0   8532  4568 ?        Ss   15:17   0:00 @dbus-daemon --system --address=systemd: --nofork --nopidfile --systemd-activation --sysl
root         185  0.0  0.2  30092 19812 ?        Ss   15:17   0:00 /usr/bin/python3 /usr/bin/networkd-dispatcher --run-startup-triggers
syslog       186  0.0  0.0 222404  7136 ?        Ssl  15:17   0:00 /usr/sbin/rsyslogd -n -iNONE
root         190  0.0  0.0  15332  7228 ?        Ss   15:17   0:00 /lib/systemd/systemd-logind
root         218  0.0  0.0   3240  1064 hvc0     Ss+  15:17   0:00 /sbin/agetty -o -p -- \u --noclear --keep-baud console 115200,38400,9600 vt220
root         220  0.0  0.1  15436  8724 ?        Ss   15:17   0:00 sshd: /usr/sbin/sshd -D [listener] 0 of 10-100 startups
root         221  0.0  0.0   3196  1068 tty1     Ss+  15:17   0:00 /sbin/agetty -o -p -- \u --noclear tty1 linux
root         232  0.0  0.2 107164 21492 ?        Ssl  15:17   0:00 /usr/bin/python3 /usr/share/unattended-upgrades/unattended-upgrade-shutdown --wait-for-si
root         301  0.0  0.0   2484   112 ?        Ss   15:17   0:00 /init
root         302  0.0  0.0   2500   120 ?        R    15:17   0:00 /init
shinu        303  0.0  0.0   6216  5268 pts/0    Ss   15:17   0:00 -bash
root         304  0.0  0.0   7528  4992 pts/1    Ss   15:17   0:00 /bin/login -f
shinu        371  0.0  0.1  16912  9136 ?        Ss   15:17   0:00 /lib/systemd/systemd --user
shinu        383  0.0  0.0 168816  3424 ?        S    15:17   0:00 (sd-pam)
shinu        404  0.0  0.0   6236  5120 pts/1    S+   15:17   0:00 -bash
shinu        721  0.0  0.0   7488  3080 pts/0    R+   16:07   0:00 ps aux
```

2. Open top command and monitor CPU usage.

```
shinu@Shinu:~$ top
top - 16:11:17 up 53 min,  1 user,  load average: 0.00, 0.00, 0.00
Tasks:  24 total,   1 running,  23 sleeping,   0 stopped,   0 zombie
%Cpu(s):  0.0 us,  0.0 sy,  0.0 ni,100.0 id,  0.0 wa,  0.0 hi,  0.0 si,  0.0 st
MiB Mem :   7642.3 total,   6990.4 free,    456.2 used,    195.8 buff/cache
MiB Swap:   2048.0 total,   2048.0 free,      0.0 used.   6959.4 avail Mem

  PID USER      PR  NI    VIRT    RES    SHR S  %CPU  %MEM     TIME+ COMMAND
    1 root      20   0  165764  11228   8316 S   0.0   0.1   0:00.96 systemd
    2 root      20   0    2476   1432   1320 S   0.0   0.0   0:00.01 init-systemd(Ub
    6 root      20   0    2492    132    132 S   0.0   0.0   0:00.00 init
   62 root      19  -1   64132  14536  13508 S   0.0   0.2   0:00.43 systemd-journal
   82 root      20   0   21960   5660   4424 S   0.0   0.1   0:00.77 systemd-udevd
  131 systemd+  20   0   26200  13456   8472 S   0.0   0.2   0:00.26 systemd-resolve
  135 systemd+  20   0   89364   6684   5880 S   0.0   0.1   0:00.31 systemd-timesyn
  178 root      20   0    4308   2744   2516 S   0.0   0.0   0:00.02 cron
  180 message+  20   0    8532   4568   3984 S   0.0   0.1   0:00.20 dbus-daemon
  185 root      20   0   30092  19812  10364 S   0.0   0.3   0:00.18 networkd-dispat
  186 syslog    20   0  222404   7136   4268 S   0.0   0.1   0:00.05 rsyslogd
  190 root      20   0   15332   7228   6284 S   0.0   0.1   0:00.21 systemd-logind
  218 root      20   0    3240   1064    976 S   0.0   0.0   0:00.00 agetty
  220 root      20   0   15436   8724   7152 S   0.0   0.1   0:00.02 sshd
  221 root      20   0    3196   1068    980 S   0.0   0.0   0:00.00 agetty
  232 root      20   0  107164  21492  12972 S   0.0   0.3   0:00.13 unattended-upgr
  301 root      20   0    2484    112      0 S   0.0   0.0   0:00.00 SessionLeader
  302 root      20   0    2500    120      0 S   0.0   0.0   0:00.14 Relay(303)
  303 shinu     20   0    6216   5268   3412 S   0.0   0.1   0:00.28 bash
  304 root      20   0    7528   4992   4068 S   0.0   0.1   0:00.01 login
  371 shinu     20   0   16912   9136   7700 S   0.0   0.1   0:00.12 systemd
  383 shinu     20   0  168816   3424     20 S   0.0   0.0   0:00.00 (sd-pam)
  404 shinu     20   0    6236   5120   3432 S   0.0   0.1   0:00.04 bash
  735 shinu     20   0    7792   3720   3120 R   0.0   0.0   0:00.03 top
```

3. Check disk usage.

```
shinu@Shinu:~$ df -h
Filesystem      Size  Used Avail Use% Mounted on
none            3.8G  4.0K  3.8G   1% /mnt/wsl
drivers         476G  393G   83G  83% /usr/lib/wsl/drivers
none            3.8G     0  3.8G   0% /usr/lib/modules
none            3.8G     0  3.8G   0% /usr/lib/modules/5.15.153.1-microsoft-standard-WSL2
/dev/sdc       1007G  6.2G  950G   1% /
none            3.8G   88K  3.8G   1% /mnt/wslg
none            3.8G     0  3.8G   0% /usr/lib/wsl/lib
rootfs          3.8G  2.1M  3.8G   1% /init
none            3.8G  528K  3.8G   1% /run
none            3.8G     0  3.8G   0% /run/lock
none            3.8G     0  3.8G   0% /run/shm
tmpfs           4.0M     0  4.0M   0% /sys/fs/cgroup
none            3.8G   96K  3.8G   1% /mnt/wslg/versions.txt
none            3.8G   96K  3.8G   1% /mnt/wslg/doc
C:\             476G  393G   83G  83% /mnt/c
```

4. Check memory usage.

```
shinu@Shinu:~$ free -m
               total        used        free      shared  buff/cache   available
Mem:            7642         450        6995           2         195        6964
Swap:           2048           0        2048
shinu@Shinu:~$
```

5. Difference between ps and top

ps command:

It shows the list of running processes at that moment only.

top command:

It shows the running processes in real time.

The information keeps updating and we can monitor CPU and memory usage continuously.

```
shinu@Shinu:~$ ps
    PID TTY          TIME CMD
    303 pts/0    00:00:00 bash
    749 pts/0    00:00:00 ps
shinu@Shinu:~$ top
top - 16:13:31 up 56 min,  1 user,  load average: 0.00, 0.00, 0.00
Tasks:  24 total,   1 running,  23 sleeping,   0 stopped,   0 zombie
%Cpu(s):  0.0 us,  0.0 sy,  0.0 ni,100.0 id,  0.0 wa,  0.0 hi,  0.0 si,  0.0 st
MiB Mem :   7642.3 total,   6992.5 free,    453.9 used,    195.9 buff/cache
MiB Swap:   2048.0 total,   2048.0 free,      0.0 used.   6961.6 avail Mem

    PID USER      PR  NI    VIRT    RES    SHR S  %CPU  %MEM     TIME+ COMMAND
      1 root      20   0  165764  11228   8316 S   0.0   0.1   0:00.98 systemd
      2 root      20   0    2476   1432   1320 S   0.0   0.0   0:00.01 init-systemd(Ub
      6 root      20   0    2492    132    132 S   0.0   0.0   0:00.00 init
     62 root      19  -1   64132  14536  13508 S   0.0   0.2   0:00.44 systemd-journal
     82 root      20   0   21960   5660   4424 S   0.0   0.1   0:00.82 systemd-udevd
    131 systemd+  20   0   26200  13456   8472 S   0.0   0.2   0:00.26 systemd-resolve
    135 systemd+  20   0   89364   6684   5880 S   0.0   0.1   0:00.32 systemd-timesyn
    178 root      20   0    4308   2744   2516 S   0.0   0.0   0:00.02 cron
    180 message+  20   0    8532   4568   3984 S   0.0   0.1   0:00.21 dbus-daemon
    185 root      20   0   30092  19812  10364 S   0.0   0.3   0:00.18 networkd-dispat
    186 syslog    20   0  222404   7136   4268 S   0.0   0.1   0:00.05 rsyslogd
    190 root      20   0   15332   7228   6284 S   0.0   0.1   0:00.22 systemd-logind
    218 root      20   0    3240   1064    976 S   0.0   0.0   0:00.00 agetty
    220 root      20   0   15436   8724   7152 S   0.0   0.1   0:00.02 sshd
    221 root      20   0    3196   1068    980 S   0.0   0.0   0:00.00 agetty
    232 root      20   0  107164  21492  12972 S   0.0   0.3   0:00.13 unattended-upgr
    301 root      20   0    2484    112      0 S   0.0   0.0   0:00.00 SessionLeader
    302 root      20   0    2500    120      0 S   0.0   0.0   0:00.15 Relay(303)
    303 shinu     20   0    6216   5268   3412 S   0.0   0.1   0:00.30 bash
    304 root      20   0    7528   4992   4068 S   0.0   0.1   0:00.01 login
    371 shinu     20   0   16912   9136   7700 S   0.0   0.1   0:00.13 systemd
    383 shinu     20   0  168816   3424     20 S   0.0   0.0   0:00.00 (sd-pam)
    404 shinu     20   0    6236   5120   3432 S   0.0   0.1   0:00.04 bash
    750 shinu     20   0    7792   3724   3120 R   0.0   0.0   0:00.00 top
```

6. What does df -h show?

```
shinu@Shinu:~$ df -h
Filesystem      Size  Used Avail Use% Mounted on
none            3.8G  4.0K  3.8G   1% /mnt/wsl
drivers         476G  393G   83G  83% /usr/lib/wsl/drivers
none            3.8G     0  3.8G   0% /usr/lib/modules
none            3.8G     0  3.8G   0% /usr/lib/modules/5.15.153.1-microsoft-standard-WSL2
/dev/sdc       1007G  6.2G  950G   1% /
none            3.8G   92K  3.8G   1% /mnt/wslg
none            3.8G     0  3.8G   0% /usr/lib/wsl/lib
rootfs          3.8G  2.1M  3.8G   1% /init
none            3.8G  528K  3.8G   1% /run
none            3.8G     0  3.8G   0% /run/lock
none            3.8G     0  3.8G   0% /run/shm
tmpfs           4.0M     0  4.0M   0% /sys/fs/cgroup
none            3.8G   96K  3.8G   1% /mnt/wslg/versions.txt
none            3.8G   96K  3.8G   1% /mnt/wslg/doc
C:\             476G  393G   83G  83% /mnt/c
```

df -h shows the disk usage of the system in human readable format.

It displays information such as:

- Total disk space
- Used space
- Available space
- Mounted file systems

Assignment 6: Shell Scripting (Basic)

Task 1: Simple Script

Write a shell script that:

- Prints “Welcome to Linux Training”;
- Prints current date
- Prints current user name

For command to print we use $(command)

```
shinu@Shinu:/mnt/c/Users/shrin/desktop/nucleusteqtraining$ touch script.sh
shinu@Shinu:/mnt/c/Users/shrin/desktop/nucleusteqtraining$ ls -l
total 0
-rwxrwxrwx 1 shinu shinu 0 Mar 14 16:32 script.sh
shinu@Shinu:/mnt/c/Users/shrin/desktop/nucleusteqtraining$ echo 'echo "Welcome to Linux Training"' >> script.sh
shinu@Shinu:/mnt/c/Users/shrin/desktop/nucleusteqtraining$ echo 'echo "Current date : $(date)"' >> script.sh
shinu@Shinu:/mnt/c/Users/shrin/desktop/nucleusteqtraining$ echo 'echo "Current User : $(whoami)"' >> script.sh
shinu@Shinu:/mnt/c/Users/shrin/desktop/nucleusteqtraining$ cat script.sh
echo "Welcome to Linux Training"
echo "Current date : $(date)"
echo "Current User : $(whoami)"
shinu@Shinu:/mnt/c/Users/shrin/desktop/nucleusteqtraining$ ./script.sh
Welcome to Linux Training
Current date : Sat Mar 14 16:33:12 IST 2026
Current User : shinu
shinu@Shinu:/mnt/c/Users/shrin/desktop/nucleusteqtraining$ |
```

Task 2: User Input Script

Create a script that:

- Takes user name as input
- Prints: “Hello <name> , welcome to Linux;

To print variable we use $variable_name

```
shinu@Shinu:/mnt/c/Users/shrin/desktop/nucleusteqtraining$ ls -l
total 0
-rwxrwxrwx 1 shinu shinu 78 Mar 14 16:41 script.sh
shinu@Shinu:/mnt/c/Users/shrin/desktop/nucleusteqtraining$ vi script.sh
shinu@Shinu:/mnt/c/Users/shrin/desktop/nucleusteqtraining$ cat script.sh
#!/bin/bash
echo "Enter name"
read name
echo "Hello $name , Welcome to Linux"
shinu@Shinu:/mnt/c/Users/shrin/desktop/nucleusteqtraining$ ./script.sh
Enter name
Shrinivas
Hello Shrinivas , Welcome to Linux
shinu@Shinu:/mnt/c/Users/shrin/desktop/nucleusteqtraining$ |
```

Conditional & Loop Script

Task 3: Even or Odd Program

```
shinu@Shinu:/mnt/c/Users/shrin/desktop/nucleusteqtraining$ ls -l
total 0
-rwxrwxrwx 1 shinu shinu 0 Mar 14 16:54 script.sh
shinu@Shinu:/mnt/c/Users/shrin/desktop/nucleusteqtraining$ vi script.sh
shinu@Shinu:/mnt/c/Users/shrin/desktop/nucleusteqtraining$ cat script.sh
echo "Enter Number"
read num
if (( num % 2 == 0 )); then
        echo "Number is even"
else
        echo "Number is odd"
fi
shinu@Shinu:/mnt/c/Users/shrin/desktop/nucleusteqtraining$ ./script.sh
Enter Number
2
Number is even
shinu@Shinu:/mnt/c/Users/shrin/desktop/nucleusteqtraining$ ./script.sh
Enter Number
3
Number is odd
```

Write a script that:

- Takes a number as input
- Checks whether number is lesser than 10 or not

```
shinu@Shinu:/mnt/c/Users/shrin/desktop/nucleusteqtraining$ ls -l
total 0
-rwxrwxrwx 1 shinu shinu 195 Mar 14 16:52 script.sh
shinu@Shinu:/mnt/c/Users/shrin/desktop/nucleusteqtraining$ vi script.sh
shinu@Shinu:/mnt/c/Users/shrin/desktop/nucleusteqtraining$ cat script.sh
echo "Enter a Number"
read num
if [ $num -lt 10 ]; then
        echo "Number is less than 10"
elif [ $num -eq 10 ]; then
        echo "Number is equal to 10"
else
        echo "Number is equal or greater than 10"
fi
shinu@Shinu:/mnt/c/Users/shrin/desktop/nucleusteqtraining$ ./script.sh
Enter a Number
12
Number is equal or greater than 10
shinu@Shinu:/mnt/c/Users/shrin/desktop/nucleusteqtraining$ ./script.sh
Enter a Number
10
Number is equal to 10
shinu@Shinu:/mnt/c/Users/shrin/desktop/nucleusteqtraining$ ./script.sh
Enter a Number
3
Number is less than 10
shinu@Shinu:/mnt/c/Users/shrin/desktop/nucleusteqtraining$
```

Task 4: Loop Program

Write a script that:

- **Prints numbers from 1 to 10**

```
shinu@Shinu:/mnt/c/Users/shrin/desktop/nucleusteqtraining$ ls -l
total 0
-rwxrwxrwx 1 shinu shinu 0 Mar 14 16:58 script.sh
shinu@Shinu:/mnt/c/Users/shrin/desktop/nucleusteqtraining$ vi script.sh
shinu@Shinu:/mnt/c/Users/shrin/desktop/nucleusteqtraining$ cat script.sh
for i in {1..10}; do
        echo "$i"
done
shinu@Shinu:/mnt/c/Users/shrin/desktop/nucleusteqtraining$ ./script.sh
1
2
3
4
5
6
7
8
9
10
```

- **Prints multiplication table of 5**

$(( )) used to perform arithmetic operations in the shell script

```
shinu@Shinu:/mnt/c/Users/shrin/desktop/nucleusteqtraining$ ls -l
total 0
-rwxrwxrwx 1 shinu shinu 49 Mar 14 17:04 script.sh
shinu@Shinu:/mnt/c/Users/shrin/desktop/nucleusteqtraining$ vi script.sh
shinu@Shinu:/mnt/c/Users/shrin/desktop/nucleusteqtraining$ cat script.sh
for i in {1..10}; do
        echo "5 * $i = $((5 * i))"
done
shinu@Shinu:/mnt/c/Users/shrin/desktop/nucleusteqtraining$ ./script.sh
5 * 1 = 5
5 * 2 = 10
5 * 3 = 15
5 * 4 = 20
5 * 5 = 25
5 * 6 = 30
5 * 7 = 35
5 * 8 = 40
5 * 9 = 45
5 * 10 = 50
shinu@Shinu:/mnt/c/Users/shrin/desktop/nucleusteqtraining$
```

Assignment 7: Mini Project

Create a script that:

1. Creates a directory backup
2. Takes current date and time
3. Copies all .txt files into backup folder
4. Prints “Backup Completed”;
5. If backup folder already exists, print “Directory already exists”;

```
shinu@Shinu:/mnt/c/Users/shrin/desktop/nucleusteqtraining$ ls -l
total 0
-rwxrwxrwx 1 shinu shinu 0 Mar 14 19:17 script.sh
shinu@Shinu:/mnt/c/Users/shrin/desktop/nucleusteqtraining$ vi script.sh
shinu@Shinu:/mnt/c/Users/shrin/desktop/nucleusteqtraining$ cat script.sh
#!/bin/bash

curr_dir=$(date +%Y%m%d_%H%M)

if [ -d "backup/$curr_dir" ]; then
        echo "Directory already exists"
else
        mkdir -p backup/$curr_dir
        cp *.txt backup/$curr_dir
        echo "Backup Completed"
fi
shinu@Shinu:/mnt/c/Users/shrin/desktop/nucleusteqtraining$ echo "This is first file" > file1.txt
shinu@Shinu:/mnt/c/Users/shrin/desktop/nucleusteqtraining$ echo "This is second file" > file2.txt
shinu@Shinu:/mnt/c/Users/shrin/desktop/nucleusteqtraining$ echo "This is third file" > file3.txt
shinu@Shinu:/mnt/c/Users/shrin/desktop/nucleusteqtraining$ ls -l
total 0
-rwxrwxrwx 1 shinu shinu  19 Mar 14 19:19 file1.txt
-rwxrwxrwx 1 shinu shinu  20 Mar 14 19:19 file2.txt
-rwxrwxrwx 1 shinu shinu  19 Mar 14 19:19 file3.txt
-rwxrwxrwx 1 shinu shinu 227 Mar 14 19:18 script.sh
```

```
shinu@Shinu:/mnt/c/Users/shrin/desktop/nucleusteqtraining$ ./script.sh
Backup Completed
shinu@Shinu:/mnt/c/Users/shrin/desktop/nucleusteqtraining$ tree
.
├── backup
│   └── 20260314_1919
│       ├── file1.txt
│       ├── file2.txt
│       └── file3.txt
├── file1.txt
├── file2.txt
├── file3.txt
└── script.sh

2 directories, 7 files
shinu@Shinu:/mnt/c/Users/shrin/desktop/nucleusteqtraining$ ./script.sh
Directory already exists
shinu@Shinu:/mnt/c/Users/shrin/desktop/nucleusteqtraining$ ./script.sh
Backup Completed
shinu@Shinu:/mnt/c/Users/shrin/desktop/nucleusteqtraining$ tree
.
├── backup
│   ├── 20260314_1919
│   │   ├── file1.txt
│   │   ├── file2.txt
│   │   └── file3.txt
│   └── 20260314_1920
│       ├── file1.txt
│       ├── file2.txt
│       └── file3.txt
│
```

- The script first takes the current date and time and uses it as the name of the backup folder inside the backup directory.
- It checks if that folder already exists. If it exists, it prints “Directory already exists” otherwise it creates the folder.
- After creating the folder the script copies all .txt files into that folder and prints “Backup Completed”.